DO NOW — RECALL

Complexity and algorithm choice

Name: _____ Class: _____ Date: _____

1. Why is a queue suitable for a print server?

2. What condition must the array satisfy?

3. How many times does a for-loop from i=0 to i<6 execute?

4. A loop of length 4 inside another loop of length 4: how many inner operations?

DO NOW — CHECK / OBJECTIVES

Correct and improve

Check page 1 in GREEN PEN. Write corrections in full.

1. Model answer

It processes jobs in arrival order (FIFO).

2. Model answer

It must be sorted.

3. Model answer

Six times.

4. Model answer

16.

Vocabulary — know these terms

Big O — Asymptotic growth of algorithm work.

O(1) — Constant growth.

O(log n) — Logarithmic growth, e.g. halving search intervals.

O(n) — Linear growth.

$O(n^2)$ — Quadratic growth, e.g. complete nested scans.

NEW LEARNING

Read, annotate and explain

Start with n

n is the number of items. More items usually mean more operations.

Trace loop nesting

One full scan performs n steps. A nested n-by-n scan performs n^2 steps.

Compare growth

When n doubles, linear work doubles; quadratic work roughly quadruples. Big O describes growth, not seconds.

Explain one step in your own words, using the correct technical terms.

I DO — WORKED EXAMPLE

Follow the worked method

Compare operation counts of one loop and two nested loops for n=5.

Predict

One loop visits five elements, so does five operations.

Explain

The nested loop repeats five operations five times, making 25.

WE DO — GUIDED PRACTICE

Try → Discuss → Improve

YOU DO — INDEPENDENT PRACTICE

Questions · Part 1

Answer independently. Show code, calculations, a trace or justified reasoning as appropriate.

Question 1

Give the Big O of a single full scan of n elements.

Question 2

For n=100, give iterations of a linear n loop and nested n×n loop.

YOU DO — INDEPENDENT PRACTICE

Questions · Part 2

Answer independently. Show code, calculations, a trace or justified reasoning as appropriate.

Question 3

Why may linear search be preferable for a single search of a short unsorted array?

EXIT TICKET / REFLECTION

Check your understanding

Complete the actual lesson Exit Ticket before checking the answer.

$O(n^2)$ means an algorithm takes n^2 seconds.

How confident are you? Circle one: Confident / Getting there / Need help

Explain your choice, identifying one concept you can now use independently.

GREEN PEN — Correct and explain

Choose one mistake from your independent work. Write the correct method and explain precisely why your original answer was wrong.

Next step: revisit the model or a relevant revision-guide section if needed.